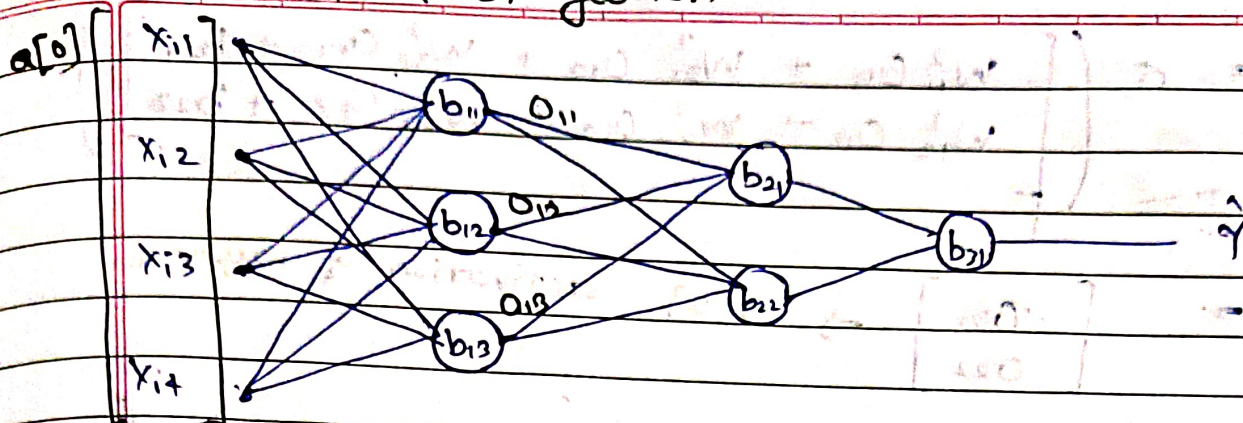


Forward Propagation



Layer 1

$$\begin{bmatrix} w'_{11} & w'_{12} & w'_{13} \\ w'_{21} & w'_{22} & w'_{23} \\ w'_{31} & w'_{32} & w'_{33} \\ w'_{41} & w'_{42} & w'_{43} \end{bmatrix}^T \begin{bmatrix} x_{11} \\ x_{12} \\ x_{13} \\ x_{14} \end{bmatrix} + \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix}$$

$$= \begin{bmatrix} w'_{11} x_{11} + w'_{21} x_{12} + w'_{31} x_{13} + w'_{41} x_{14} \\ w'_{12} x_{11} + w'_{22} x_{12} + w'_{32} x_{13} + w'_{42} x_{14} \\ w'_{13} x_{11} + w'_{23} x_{12} + w'_{33} x_{13} + w'_{43} x_{14} \end{bmatrix} + \begin{bmatrix} b_{11} \\ b_{12} \\ b_{13} \end{bmatrix}$$

$$= \begin{bmatrix} w'_{11} x_{11} + w'_{21} x_{12} + w'_{31} x_{13} + w'_{41} x_{14} + b_{11} \\ w'_{12} x_{11} + w'_{22} x_{12} + w'_{32} x_{13} + w'_{42} x_{14} + b_{12} \\ w'_{13} x_{11} + w'_{23} x_{12} + w'_{33} x_{13} + w'_{43} x_{14} + b_{13} \end{bmatrix}$$

$$= \begin{bmatrix} 0.11 \\ 0.12 \\ 0.13 \end{bmatrix} \Rightarrow a[1]$$

Layer 2

$$\begin{bmatrix} w^2_{11} & w^2_{12} \\ w^2_{21} & w^2_{22} \\ w^2_{31} & w^2_{32} \end{bmatrix}^T \begin{bmatrix} 0.11 \\ 0.12 \\ 0.13 \end{bmatrix} + \begin{bmatrix} b_{21} \\ b_{22} \end{bmatrix}$$

$$3 \times 2 \rightarrow 2 \times 1$$

$$= \sigma \left(\begin{bmatrix} w_{11}^2 o_{11} + w_{21}^2 o_{12} + w_{31}^2 o_{13} + b_{21} \\ w_{12}^2 o_{11} + w_{22}^2 o_{12} + w_{32}^2 o_{13} + b_{22} \end{bmatrix} \right)$$

$$= \begin{bmatrix} o_{21} \\ o_{22} \end{bmatrix} \Rightarrow a[2] \quad \text{activation of layer 2}$$

layer 3

$$= \begin{bmatrix} w_{11}^3 \\ w_{21}^3 \end{bmatrix}^T \begin{bmatrix} o_{21} \\ o_{22} \end{bmatrix} + \begin{bmatrix} b_{31} \end{bmatrix}$$

$$= \sigma \left(\begin{bmatrix} w_{11}^3 o_{21} + w_{21}^3 o_{22} + b_{31} \end{bmatrix} \right)$$

$$= o_{31} \Rightarrow \hat{y}_1 \Rightarrow a[3]$$

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$$\therefore \sigma \left(\sigma \left(\left(\sigma \left(a[0] w^{[1]} + b^{[1]} \right) \right) w^{[2]} + b^{[2]} \right) \right) w^{[3]} + b^{[3]}$$

$$\underbrace{\sigma \left(\sigma \left(\left(\sigma \left(a[0] w^{[1]} + b^{[1]} \right) \right) w^{[2]} + b^{[2]} \right) \right)}_{a[2]}$$

$$\underbrace{\left(\left(\left(\left(\left(\left(a[0] w^{[1]} + b^{[1]} \right) w^{[2]} + b^{[2]} \right) w^{[3]} + b^{[3]} \right) \right) \right) \right)}_{a[3]}$$

$$a[3] \Rightarrow \hat{y}$$

$$\begin{bmatrix} 1.0 \\ 0.0 \\ 0.0 \end{bmatrix}$$